

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	Total Marks:	100
C03 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	Shahana	Reg. No.:	192524479
Section / Batch:	Slot A	Date:	29/07/2026

Code of Conduct

I Shahana (192524479) _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Shahana _____

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
GRAND TOTAL:		100 Marks	

Annexure A – Interactive Tree Simulator

- Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80
Perform:
 - Inorder Traversal
 - Search for 60
 - Delete 30
 - Display the final BST.
- Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
 - Inorder
 - Preorder
 - Postorder
- Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
- Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
- Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. BST

a. INORDER TRAVERSAL

Input
Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

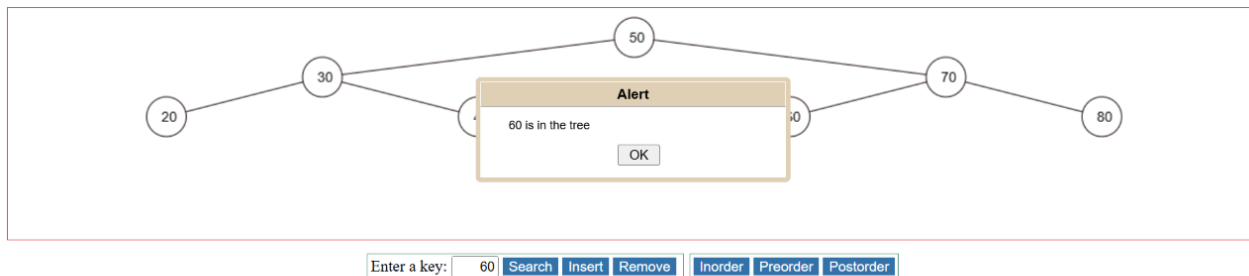
20 30 40 50 60 70 80

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80)); style 80 fill:#f96
```

b. SEARCH

BST [Animation](#) by [Y. Daniel Liang](#)

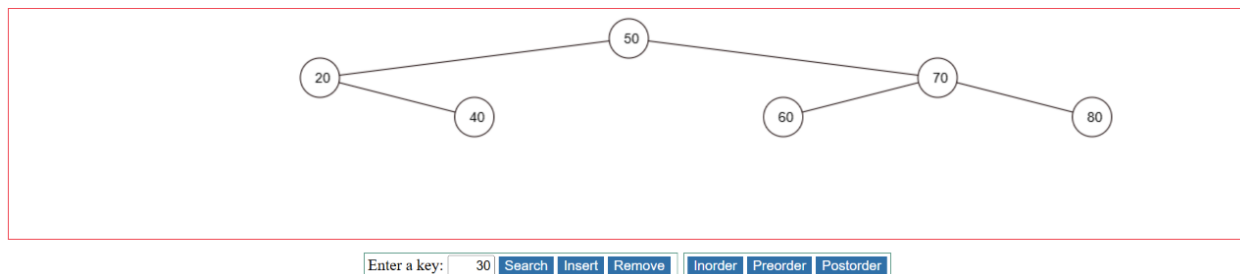
Usage: Enter an integer key and click the Search button to search the key in the tree. Click the Insert button to insert the key into the tree. Click the Remove button to remove the key from the tree. For the best display, use integers between 0 and 99. You can also display the elements in inorder, preorder, and postorder.



c. DELETE

BST [Animation](#) by [Y. Daniel Liang](#)

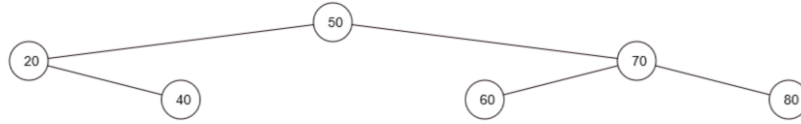
Usage: Enter an integer key and click the Search button to search the key in the tree. Click the Insert button to insert the key into the tree. Click the Remove button to remove the key from the tree. For the best display, use integers between 0 and 99. You can also display the elements in inorder, preorder, and postorder.



d. FINAL BST

BST Animation by Y. Daniel Liang

Usage: Enter an integer key and click the Search button to search the key in the tree. Click the Insert button to insert the key into the tree. Click the Remove button to remove the key from the tree. For the best display, use integers between 0 and 99. You can also display the elements in inorder, preorder, and postorder.



Enter a key:

2. TREE TRAVERSALS

a. INORDER

Input
Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

20 30 40 50 60 70 80

```
graph TD; 50((50)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

b. PREORDER

Input
Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

50 30 20 40 70 60 80

```
graph TD; 50((50)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

c. POSTORDER

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

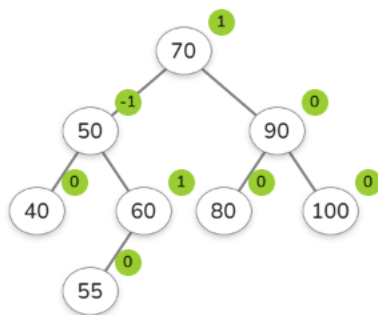
Generate Default

Visualize

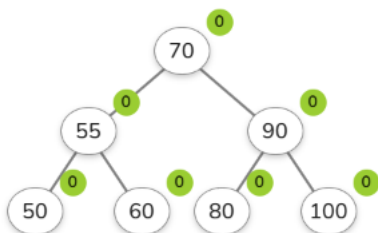
Start Simulation

20 40 30 60 80 70 50

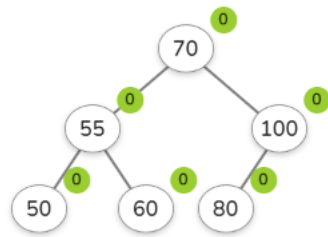
3. AVL TREE



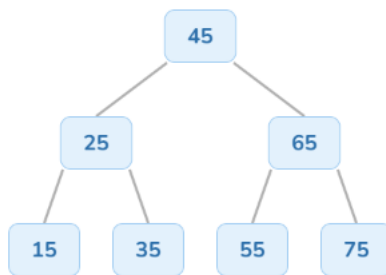
DELETE 40



DELETE 90



4. B- TREE



5. RED- BLACK TREE

